

Spherical Fourier-Bessel Clustering Analysis of the Lyman- α Forest

Roger de Belsunce,  Ant3n Baleato Lizancos  Martin White 

Berkeley Center for Cosmological Physics, UC Berkeley, CA 94720, USA

Lawrence Berkeley National Laboratory, One Cyclotron Road, Berkeley, CA 94720, USA

E-mail: rbelsunce@berkeley.edu, a.baleatolizancos@berkeley.edu,
mwhite@berkeley.edu

Abstract. We present a novel estimator for computing the three-dimensional power spectrum of the Lyman- α (Ly- α) forest flux fluctuations in the spherical Fourier-Bessel (sFB) framework. The algorithm efficiently exploits the sparse sampling transverse to and dense sampling along the line-of-sight, typical for Ly- α surveys. The decomposition in spherical harmonics and the coverage of the Ly- α forest for a comoving distance along the line-of-sight enables an efficient measurement of unequal-time angular power spectra; effectively measuring $C_\ell(k)$. This basis allows to isolate (and marginalize out) modes that are, e.g., sensitive to distortions in the continuum fitting (purely transverse modes) and modes that are sensitive to systematics that slowly vary across the sky (almost purely radial modes). Additionally, the angular power spectra are calculated based on angles on the sky, making them cosmology independent and removing the need to correct for the so-called Alcock-Paczynski effect. We present extensive tests on idealized Gaussian random field simulations with a non-trivial survey geometry. We find that the sFB estimator is robust to survey geometries and can be used to accurately measure the angular power spectrum of Ly- α forest flux fluctuations. We show a promising new approach for auto- and cross-spectrum based cosmological analyses of Ly- α forest data collected with the currently observing Dark Energy Spectroscopic Instrument and future surveys such as the Prime Focus Spectrograph, WEAVE-QSO and 4MOST. We make public a GPU-based code enabling the fast application of our method to existing and upcoming datasets. The code scales as $\mathcal{O}(N^2)$ for N quasars, taking ~ 1 second to compute the angular power spectrum per bin along the line-of-sight for 10^7 quasars.

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Notes

data is at NERSC at: [/pscratch/sd/r/rmvd2/DESI/iron/deltas_lya/Delta/DESI_iron_all_lya.fits](#) save as one large dictionary with two marices: (i) for the weights, ie the window; (ii) for the data, ie the deltas.

[MW: Do we need to deal with eliminating the auto-spectra? We could subtract this from the estimator, or include it in the model.] [ABL: Some of these contractions give us the P_{1D} , don't they? If so, I feel that's valuable information that I think we should model. But I'm probably missing something.] [MW: No, I think you're right that the "diagonal" or "auto-spectrum" contribution is P_{1D} .] [RB: Is $C_{\ell=0}(k)$ the P_{1D} ?]

[RB: Ideally we would include auto-spectra / self-skewer contributions since this is additional information. So far, these were removed to account for continuum-uncertainty-type issues. For simulations we can just include them for real data we'll have to see the magnitude of the effect...]

1 Introduction

We propose a novel class of estimators expanding the harmonic-space analysis approach of the cosmic microwave background (CMB) and galaxy clustering (see, e.g., [1–3]) to the 3D analysis of Lyman- α (Ly- α) forest flux fluctuations. This series of absorption features stems from neutral hydrogen in the low-density, highly ionized intergalactic medium (IGM), observed in quasar spectra [4]. These absorption patterns are a measure of the distribution of neutral hydrogen along the line-of-sight. They are a powerful probe at intermediate redshifts ($2 \lesssim z \lesssim 5$) of the large-scale structure of the Universe at Mpc scales and below. The Ly- α forest provides constraints on cosmological parameters through the BAO feature [5–8], broadband shape of the 3D Ly- α correlations (see, e.g., [9]), neutrino masses [10–12], primordial black holes [13], the thermal properties of the ionized (cold) IGM (see, e.g., [14]) and can discriminate between warm dark matter models (see, e.g., [15, 16]).

A key challenge when performing cosmological inference from Ly- α forest spectra is, that quasars are sparse on the sky with the flux being well sampled for a range of comoving distances, χ , along each line-of-sight. This suggests a ‘radial-angular’ (e.g. spherical Fourier-Bessel) analysis of 3D fields [2, 17–24], as an alternative to the 3D flux power spectrum [25–28]. We exploit the radial-angular structure, *i.e.* a natural basis, of the Ly- α forest. Our measurement is reported as $C_\ell(k)$, allowing for tomographic analyses of the Ly- α forest¹. This paves the way for auto-correlation analyses to constrain early-Universe physics such as the amplitude and the tilt of the primordial power spectrum and cross-correlation analyses of different redshift (or comoving) bins or correlations with long-wavelength tracers, e.g. CMB lensing, to constrain the amplitude of matter clustering in a novel redshift regime (see, e.g., [29]).

The two-point correlation function (2PCF) and the power spectrum, $P(k)$, are among the most commonly used summary statistics to study the large-scale structure (LSS) of the Universe. While the 2PCF is a powerful tool to study effects arising at specific separations, *i.e.* the baryonic acoustic oscillation (BAO) feature, the power spectrum is able to isolate slowly varying effects down to large scales. However, both approaches are defined in terms of *cosmology-dependent* variables; giving rise to redshift-space distortions and the (so-called) Alcock-Paczynski effect [30]. Manifesting themselves as distortions in the galaxy distribution with anisotropies around the line-of-sight. These arise if an incorrect fiducial cosmological model is assumed when reconstructing 3D positions of tracers from the naturally observed (*cosmology independent*) redshift and angular positions on the sky. Therefore, harmonic-space estimators have been widely adopted in the context of CMB and galaxy clustering analyses (see, e.g., [17, 18]). They enable auto- and cross-spectrum analyses of (different) projected scalar fields on the sphere. The advent of the currently observing Dark Energy Spectroscopic Instrument [DESI; 31] and next-generation experiments such as Euclid [32], SPHEREx [33], Rubin [34], and Roman [35] will map the Universe to unprecedented precision and require accurate and fast estimators to extract the cosmological information at hand.

An additional complexity when computing summary statistics of the Ly- α forest stems from the window matrix due to highly anisotropic selection function [27]. We forward model the window function to avoid instabilities stemming from the inversion of the sparse, non-local window matrix. Therefore, when comparing measured angular power spectra, $\hat{C}_\ell(k)$, to theory spectra, $C_\ell(k)$, we apply the window function to the theory.

The remainder of this paper is organized as follows: We present the angular pseudo-power spectrum estimator in Sec. 2 together with the forward modeling of the window function. In Sec. 3 we present tests of our pipeline on idealized Gaussian random field simulations and Sec. 4 presents our conclusions.

Our `directSHT-lya` implementation is publicly available.²

2 The spherical Fourier-Bessel angular pseudo-power spectrum

Let us consider an arbitrary projected field

$$\delta_{2D}(\hat{n}; k) = \int d\chi K(\hat{n}, \chi; k) \delta_{3D}(\chi \hat{n}) , \quad (2.1)$$

¹Analogous arguments apply for any tracer observed on the sky with an associated redshift.

²<https://github.com/roger2b/directSHT-lya>

where for the Ly- α forest the particular form of interest will have $\delta_{3D} = \delta_F(\chi \hat{n})$ and³, χ the comoving distance along the line-of-sight, $\hat{n}$ the unit vector in the direction of the line-of-sight, and k the wavenumber. The kernel K defines the 3D mask of the observed Ly- α forest field

$$K(\hat{n}, \chi; k) = \sum_j \delta(\hat{n} - \hat{n}_j) K_j(\chi) e^{ik\chi} \Rightarrow K_{\ell m}(\chi; k) = \sum_j Y_{\ell m}^*(\hat{n}_j) K_j(\chi) e^{ik\chi} \quad (2.2)$$

where we have also indicated the spherical harmonic transform, $K_{\ell m}(\chi; k)$. Note that the χ integral corresponds to performing a Fourier (not Hankel) transform in the line of sight direction, with a weighting function $K_j(\chi)$ for sightline j . This will be followed by a harmonic transform for each k mode which is only sampled at the positions of the QSOs. A typical convention would be to give $K_j(\chi)$ dimensions of inverse length such that its Fourier transform is dimensionless. We further define

$$K_{\ell}(\chi_1, \chi_2; k) \equiv \frac{1}{2\ell + 1} \sum_m K_{\ell m}(\chi_1; k) K_{\ell m}^*(\chi_2; k) \quad (2.3)$$

$$= \frac{1}{2\ell + 1} \sum_{jkm} Y_{\ell m}^*(\hat{n}_j) Y_{\ell m}(\hat{n}_k) K_j(\chi_1) K_k(\chi_2) e^{ik(\chi_1 - \chi_2)} \quad (2.4)$$

$$= \frac{1}{4\pi} \sum_{jk} P_{\ell}(\hat{n}_j \cdot \hat{n}_k) K_j(\chi_1) K_k(\chi_2) e^{ik(\chi_1 - \chi_2)}, \quad (2.5)$$

where $P_{\ell}(\hat{n})$ is the Legendre polynomial. The angular pseudo-power spectrum of δ_{2D} is [1, 36]

$$C_{\ell}(k) = \sum_L \int d\chi_1 d\chi_2 M_{\ell L}(\chi_1, \chi_2; k) C_L(\chi_1, \chi_2) \quad (2.6)$$

where

$$M_{\ell L} = \frac{2L + 1}{4\pi} \sum_{\lambda} (2\lambda + 1) \begin{pmatrix} \ell & L & \lambda \\ 0 & 0 & 0 \end{pmatrix}^2 K_{\lambda}(\chi_1, \chi_2; k), \quad (2.7)$$

and the multi-frequency angular power spectrum (MAPS [37]; see also refs. [22, 38, 39] and references therein)

$$C_L(\chi_1, \chi_2) = \frac{2}{\pi} \int k^2 dk P(k, \chi_1, \chi_2) j_L(k\chi_1) j_L(k\chi_2) \quad (2.8)$$

$$\approx \bar{\chi}^{-2} \int \frac{dk}{2\pi} P(k_{\perp} = L/\bar{\chi}, k_{\parallel} = k) e^{ik\Delta\chi} \quad (2.9)$$

with $\bar{\chi} = (\chi_1 + \chi_2)/2$ and $\Delta\chi = \chi_1 - \chi_2$. The approximation on the second line is valid in the “flat-sky limit”, or when $L \gg 1$, which covers all of our cases of interest. It involves only the equal-time power spectrum, rather than the unequal-time spectrum in the full expression. The approach to the flat sky limit (i.e. corrections to the above) is well understood, see for example ref. [2] for a recent discussion. **[MW: What we put for $P(k)$ in the “exact” expression is slightly ambiguous in the case of redshift space distortions – important for Ly- α – the exact expression is Eq. (26) of [22] for example.]**

³Having the δ -function in here means we can’t do a “variance” of this mask.

By recasting Eq. (2.7) to

$$F_{\ell L \lambda} = \frac{(2L+1)(2\lambda+1)}{(4\pi)^2} \begin{pmatrix} \ell & L & \lambda \\ 0 & 0 & 0 \end{pmatrix}^2 \quad (2.10)$$

we simplify the presentation and combine some of these expressions [MW: The $\bar{\chi}$ approximation worries me. I'm wondering if we want to define $\bar{\chi}$ as a fixed mid-point or something, rather than the average of the two χ_i ? Or we could imagine breaking K_j up into contributions each of which is “short” and then adding another sum over these.]

$$C_\ell(k) = \sum_{L\lambda} F_{\ell L \lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int d\chi_1 d\chi_2 K_j(\chi_1) K_k(\chi_2) e^{ik(\chi_1 - \chi_2)} C_L(\chi_1, \chi_2) \quad (2.11)$$

$$\begin{aligned} &= \sum_{L\lambda} F_{\ell L \lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int \frac{d\chi_1 d\chi_2}{\bar{\chi}^2} \int \frac{dk_1 dk_2}{(2\pi)^2} \tilde{K}_j^*(k_1) \tilde{K}_k(k_2) \\ &\times \int \frac{dk'}{2\pi} P(L/\bar{\chi}, k') e^{i(k-k'-k_1)\chi_1} e^{-i(k-k'-k_2)\chi_2} \end{aligned} \quad (2.12)$$

$$= \sum_{L\lambda} F_{\ell L \lambda} \int \frac{dk'}{2\pi \bar{\chi}^2} P(L/\bar{\chi}, k') \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \tilde{K}_j^*(k-k') \tilde{K}_k(k-k') \quad (2.13)$$

$$= \sum_{L\lambda} F_{\ell L \lambda} \int \frac{dk'}{2\pi \bar{\chi}^2} P(L/\bar{\chi}, k') \frac{4\pi}{2\lambda+1} W_\lambda(k-k') \quad (2.14)$$

$$= \sum_L \frac{2L+1}{4\pi} \int \frac{dk'}{2\pi \bar{\chi}^2} \mathcal{M}_{\ell L}(k-k') P(L/\bar{\chi}, k') \quad (2.15)$$

[MW: The pre-factor is so the $\sum_L (2L+1)/(4\pi)$ would become $\int L dL/(2\pi)$ for large L .] with the obvious definitions of W_λ and $\mathcal{M}$. In the second step we assumed that $K_j(\chi)$ is such that $\bar{\chi}$ can be taken to be constant. If the line-of-sight is long compared to the wavelength associated with k , and $K_j(\chi)$ has very little structure, then $\tilde{K}_j^* \tilde{K}_k$ has support only when $k' \simeq k$. Further, if the weighted distribution of QSOs is approximately uniform and covers a significant portion of the sky, W_λ is small for $\lambda \gg 1$. [MW: For 10^4 uniformly distributed QSOs over $1/3$ of the sky with top-hat $K_j(\chi)$, $W_\lambda < 10^{-3} W_0$ for $\lambda \geq 10$ and $W_0 < 10^{-4}$ for $\lambda \geq 20$ if we omit self-pairs and only a small factor larger if we include them.]

[MW: How do we handle self-pairs?] [RB: Ideally we would want to include them since this gives additional information. For the simulations we can just include them, for real data we'll have to see the magnitude of the effect... Can we maybe explicitly subtract self-pairs?]

If W_λ is small for $\lambda \gg 1$ and $P(L/\bar{\chi}, k')$ is slowly varying in L then we can neglect that variation and use $\sum_L F_{\ell L \lambda} = (2\lambda+1)/(4\pi)^2$. Then

$$C_\ell(k) \approx \int \frac{dk'}{2\pi \bar{\chi}^2} P(\ell/\bar{\chi}, k') \sum_\lambda \frac{W_\lambda(k-k')}{4\pi} = \int \frac{dk'}{2\pi \bar{\chi}^2} P(\ell/\bar{\chi}, k') W(k-k') \quad (2.16)$$

This pseudo-spectrum would typically be binned in ℓ (and possibly k). [MW: I'm worried about this approximation and the sum over λ .]

As a last step, we note that – while it is not required – it can be helpful to normalize $C_\ell(k)$. One normalization convention is to set $C_\ell(k) = 1$ for a shot-noise spectrum with $P = \bar{\chi}^3$. [MW: Probably not a good choice, because we can't do a “variance” of our mask!]

This amounts to multiplying Eq. (2.15) by $\mathcal{N}$ where

$$\mathcal{N}^{-1} = \frac{1}{4\pi} \sum_{\lambda} \int \frac{\bar{\chi} dk'}{2\pi} W_{\lambda}(k - k') \quad (2.17)$$

$$= \frac{1}{4\pi} \sum_{\lambda j k} \frac{2\lambda + 1}{4\pi} P_{\lambda}(\hat{n}_j \cdot \hat{n}_k) \int \frac{\bar{\chi} dk'}{2\pi} \tilde{K}_j^*(k') \tilde{K}_k(k') \quad (2.18)$$

$$= \frac{1}{4\pi} \sum_{\lambda j k} \frac{2\lambda + 1}{4\pi} P_{\lambda}(\hat{n}_j \cdot \hat{n}_k) \int \bar{\chi} d\chi K_j(\chi) K_k(\chi) \quad (2.19)$$

$$= \frac{1}{4\pi} \sum_{\lambda j k} \frac{2\lambda + 1}{4\pi} P_{\lambda}(\hat{n}_j \cdot \hat{n}_k) \mathcal{K}_{jk} \quad (2.20)$$

[MW: Now the sum over λ just a δ -function! If I include self-pairs I get $\mathcal{N} = 0$ and if I omit them $\mathcal{N}^{-1} = 0$.] which is dimensionless if $K_j(\chi)$ has dimensions of inverse length. Note if the wavelength grid of the QSO spectra contains N_{bin} bins, $\mathcal{K}_{jk}$ can be expressed as the product of an $N_{\text{qso}} \times N_{\text{bin}}$ matrix times its transpose. If the matrix $\hat{n}_j \cdot \hat{n}_k$ is pre-computed, the sum over λ can be done with recurrence relations and the final sum over jk is a matrix product and trace. For very large numbers of sightlines, the sum over j can be done in batches using the same procedure.

[MW: Perhaps a better normalization would be to set something equal to N_{qso} or $\mathcal{K}_{ii} = 1$?]

2.1 Implementation details

The computation of $C_{\ell}(k)$ can be done quite efficiently. First one computes $\delta_{2D}(\hat{n}_j; k)$ for $\delta_{3D} = \delta_F(\chi \hat{n}_j)$ along each QSO sightline, $\hat{n}_j$, either using FFTs⁴ or as a matrix-vector multiply of $\exp[ik\chi]$ with $K_j(\chi)\delta_F(\chi)$. Since the DESI QSO spectra all share a common wavelength grid, the “matrix” can be computed once. It is convenient to store the real and imaginary parts separately. The same procedure on $K_j(\chi)$ not weighted by δ_F gives $\tilde{K}_j(k)$.

Using $\delta_{2D}(\hat{n}_j; k)$ as weights for each QSO one performs a spherical harmonic transform for each k . For $\ell_{\text{max}} = 10^3$ and 10^6 QSOs this takes a few seconds for each of the real and imaginary parts for each k . Squaring the results and averaging over m gives $C_{\ell}(k)$, which can be binned into bins of ℓ and normalized as desired. If memory is not an issue, this step can be accelerated by pre-computing and storing the $Y_{\ell m}(\hat{n}_j)$.

Finally to perform inference we need to compute the window matrix. This is independent of the data, depending only upon the chosen weights, $K_j(\chi)$, and the QSO positions, $\hat{n}_j$. If the matrix $\hat{n}_j \cdot \hat{n}_k$ is pre-computed, the sum over λ can be done with recurrence relations and the final sum over jk is a matrix product and trace though the final object depends upon ℓ (normally binned), L and $k - k'$ and so can be large. For very large numbers of sightlines, the sum over j can be done in batches using the same procedure.

[MW: I’ve assumed here we’re just FFTing in the LOS direction. But per spectrum there’s not many pixels, so a full C^{-1} weighting wouldn’t be computationally intractable.]

⁴High column density systems appearing in the Ly- α forest are masked (flux zeroed out) for cosmological analyses (see, e.g. [8]). Thus, we advocate for weighting pixels by $\exp[ik\chi]$ for wavenumber k and comoving distance χ to avoid introducing aliasing due to masked regions within the spectra.

2.2 Deprojection of modes to reduce systematics

[RB: Discuss how we can measure the power spectrum in $k_{\parallel}$ and $k_{\perp}$ (or ℓ) bins which makes it ideal to de-project modes that are strongly affected by systematics: We expect these modes to be very difficult to measure in a real data analysis, since $k_{\parallel} = 0$ modes are very sensitive to distortions caused in continuum fitting and $k_{\perp} = 0$ are very sensitive to systematics that vary slowly in the sky.]

2.3 GPU implementation

[RB: Should we include this or is this a separate paper with Kendrick?]

3 Consistency Tests on Gaussian random field simulations

We test the presented angular power spectrum estimator on Gaussian random fields (GRFs) including anisotropies. This first series of tests investigates biases as well as the range of validity and robustness of the estimator to non-trivial survey geometries. Note that we ignore crucial contaminants, e.g., spectral resolution, continuum fitting, high column density absorbers and metals. We generate idealized Gaussian realizations with a fiducial linear matter power spectrum, $P_{\text{lin}}(k)$. We use the well The GRFs are set in a periodic box of size $L = 1380 h^{-1} \text{ Mpc}$ with $N = 512$ cells at redshift $z = 2.4$ with a ΛCDM linear power spectrum as theory input [40]. The fundamental mode is $k_F = 2\pi/L = 0.0045 h^{-1} \text{ Mpc}$, the maximum mode that we can reconstruct is $k_{\text{max}} = 2.02 h^{-1} \text{ Mpc}$ and the Nyquist frequency is $k_{\text{Ny}} = 1.17 h^{-1} \text{ Mpc}$.

We sample two density configurations from the GRF: First, a dense sampling mimicking a DESI-like survey $N_s = 9000$ sight lines from the box, corresponding to a DESI-like quasar density of $\sim 30 \text{ qso/deg}^2$. The computation uses $\sim 4.5 \cdot 10^6$ Ly- α pixels for the present analysis. For the data weights we use the pixel value, *i.e.*, the modes from the GRF and for the random weights we use constant weights of unity. For the power spectrum measurement we include self-skewers⁵. Additionally, we apply a non-trivial selection function, *i.e.*, with edges and sparse sampling transverse to the line-of-sight to measure the effect of the window function on the measured power spectrum. We remove the periodicity of the box and add 10% root-mean-squared error, relative to the mean, Gaussian, white (uncorrelated), noise to each of the pixels in the box, corresponding to somewhat realistic Ly- α noise levels for continuum fitting and pipeline noise. Therefore, we draw from a normal distribution $\mathcal{N}(0, \sigma_c^2)$, where σ_c denotes our uncertainty budget.

4 Summary and Conclusions

The Ly- α forest is a treasure trove of cosmological information on the expansion history of the Universe and beyond. Given its high-redshift range ($2 \leq z \leq 4$) and sensitivity to Mpc scales and below, it is an ideal tracer to probe a wide range of scales: early-Universe physics from combinations with a large-scale tracer, e.g., the cosmic microwave background anisotropies, from large scales or dark matter models as well as thermal properties of the ionized (cold) IGM from small scales. The radial-angular structure of the Ly- α forest lends itself well to a spherical Fourier-Bessel analysis and is a powerful alternative to the recently

⁵For an approach to measure the 3D power spectrum using a pair counting algorithm that includes/rejects self-skewer contributions, see [27].

measured three-dimensional power spectrum [27].⁶ In this paper, we propose a spherical harmonic analysis of the Ly- α forest to effectively measure $C_\ell(k)$.

Discuss that this estimator is in the natural basis of the Ly- α forest, similar to the optimal estimator presented in [25].

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A Beyond flat sky

[MW: If we want we could put in a first correction to the flat-sky formula presented in the main text. This is the correction term between $C_L(\chi_1, \chi_2)$ and the FT of $P(k_\perp = L/\bar{\chi}, k_\parallel)$. This would be small except at low L , so we might be able to handle this more carefully. This does, however, naturally bring in questions about mean flux evolution and unequal time correlators. The FT of a j_ℓ is a ratio of Γ -functions with complex arguments, so the expressions aren’t too difficult in principle but can be unwieldy in practice.]

[MW: I wonder if we could alternatively use the Rayleigh expansion rewritten as an integral representation of j_ℓ

$$j_\ell(x) = \frac{(-i)^\ell}{2} \int_{-1}^{+1} dk e^{ikx} P_\ell(k) = \int_{-\infty}^{+\infty} \frac{dk}{2\pi} e^{ikx} \left[\pi(-i)^\ell P_\ell(k) \Pi(k) \right] \quad (\text{A.1})$$

where $\Pi(k)$ is the “rectangle” function restricting $-1 \leq k \leq 1$. Given our $W_j(\chi)$ factors we could always redefine our χ integrals to run from -1 to 1 through e.g. $(\chi_1 - \bar{\chi})/\Delta\chi$ or something.]

[MW: What is unambiguously defined is $\xi(\triangle)$, where $\triangle$ indicates the triangle “shape” for the observer and the two objects being correlated. ξ is independent of rotation of the line-of-sight direction and around the line-of-sight direction. The $C_\ell(s_1, s_2)$ statistic corresponds to describing the triangle by the (scalar) lengths of the two sides and the enclosed angle (which then expanded in Legendre polynomials). We are then doing the 1D FFT on each of the arguments, s_i . This is very close to the sFB approach, where we’d Hankel transform on the side lengths. It’s a bit different than other approaches. We can easily work out (i.e. copy) the linear theory for $\xi(\triangle)$. I’m not sure the best formalism/approach to take.]

[MW: Section 2 of ref. [22] discusses several “coordinate systems” for describing $\xi(\triangle)$ and gives relations to the LOS-dependent power-spectrum, $P(\mathbf{k}, d)$. I’ve copied a few formula below just in case they’re useful (though they’re probably not...):]

Describing $\triangle$ by the two side lengths and enclosed angle gives the definition of $C_\ell(s_1, s_2)$:

$$\xi(\mathbf{s}_1, \mathbf{s}_2) = \sum_\ell \frac{2\ell + 1}{4\pi} C_\ell(s_1, s_2) P_\ell(\hat{\mathbf{s}}_1 \cdot \hat{\mathbf{s}}_2) \quad (\text{A.2})$$

⁶Paired with improved theoretical modeling of the Ly- α forest [see, e.g., 41] this paves the way for full-shape cosmological analyses of the Ly- α forest.

If we invert this statement

$$C_\ell(s_1, s_2) = 2\pi \int d(\cos \theta) \xi(\mathbf{s}_1, \mathbf{s}_2) P_\ell(\cos \theta) \quad (\text{A.3})$$

Then approximating $P_\ell(\cos \theta) \approx J_0(\ell \tilde{\omega})$ with $\tilde{\omega} = 2 \sin(\theta/2)$ we have

$$C_\ell(s_1, s_2) \approx 2\pi \int \tilde{\omega} d\tilde{\omega} \xi(s_\perp, s_\parallel, d) J_0(\ell \tilde{\omega}) \quad (\text{A.4})$$

Treating the J_0 as from the Rayleigh expansion of the plane wave in cylindrical coordinates, and switching to “center of mass and relative coordinates”

$$C_\ell(s_\parallel, d) \approx \int \frac{d^2 s_\perp}{d^2} \xi(s_\perp, s_\parallel, d) e^{i\mathbf{k}_\perp \cdot \mathbf{s}_\perp} \quad (\text{A.5})$$

which is the 2D FT of the correlation function. If we further do the FT in $s_\parallel$ we get $P(\mathbf{k})$, or phrased another way

$$C_\ell(s_\parallel, d) \approx \int_0^\infty \frac{dk_\parallel}{\pi d^2} P(k_\perp = \ell/d, k_\parallel) \cos(k_\parallel s_\parallel) \quad (\text{A.6})$$

The above shows that $C_\ell(s_1, s_2)$ is easy to interpret in the flat sky limit. The alternative, and the most commonly used definition in sFB, is to Hankel transform each leg:

$$C_\ell(k_1, k_2) = \int s_1^2 ds_1 \int s_2^2 ds_2 C_\ell(s_1, s_2) j_\ell(k_1 s_1) j_\ell(k_2 s_2) \quad (\text{A.7})$$

According to Eq. (26) of ref. [22] we have

$$C_\ell(s_1, s_2) = \sum_{L\lambda} G_{L\lambda}^\ell \int \frac{k^2 dk}{2\pi^2} P_L(k, s_1) j_\lambda(k s_1) j_\ell(k s_2) \quad (\text{A.8})$$

where

$$G_{L\lambda}^\ell = 4\pi(2\lambda + 1) i^{\lambda-\ell} \begin{pmatrix} \ell & L & \lambda \\ 0 & 0 & 0 \end{pmatrix}^2 \quad (\text{A.9})$$

(called F in ref. [22], but we’ve used F earlier so I’m calling it G for now – maybe some other name is better). So doing the FT along s_i is going to involve the FT of a spherical Bessel function. If we can figure out how the flat-sky limit is approached (e.g. figure out what [2] are saying) it should just drop out that we get $P(k_\perp = \ell/d, k_\parallel)$.

[RB: Maybe the difference comes from Eq. (11) in [22] where you take the observed power spectrum. In [2] they use the theory matter power spectrum. But the flat-sky approximation should still hold, I believe for Eqs. (2.17)-(2.18) from <https://arxiv.org/pdf/2307.13768.pdf>.]

B Scratch space

[MW: Some scratch space to work out some things that will probably not make it into the paper.]

Since the K_j have finite support, we can extend the integrals over χ_i to the entire real line. Defining $\bar{\chi} = (\chi_1 + \chi_2)/2$ and $\Delta\chi = \chi_1 - \chi_2$

$$C_\ell(k) = \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int d\chi_1 d\chi_2 K_j(\chi_1) K_k(\chi_2) e^{ik(\chi_1 - \chi_2)} C_L(\chi_1, \chi_2) \quad (\text{B.1})$$

$$\begin{aligned} &= \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int \frac{d\chi_1 d\chi_2}{\bar{\chi}^2} \int \frac{dk_1 dk_2}{(2\pi)^2} \tilde{K}_j^*(k_1) \tilde{K}_k(k_2) \\ &\times \int \frac{dk'}{2\pi} P(L/\bar{\chi}, k') e^{i(k-k'-k_1)\chi_1} e^{-i(k-k'-k_2)\chi_2} \end{aligned} \quad (\text{B.2})$$

$$\begin{aligned} &= \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int \frac{d\bar{\chi} d\Delta\chi}{\bar{\chi}^2} \int \frac{dk_1 dk_2}{(2\pi)^2} \tilde{K}_j^*(k_1) \tilde{K}_k(k_2) \\ &\times \int \frac{dk'}{2\pi} P(L/\bar{\chi}, k') e^{i(k-k')\Delta\chi} e^{-i(k_1\chi_1 - k_2\chi_2)} \end{aligned} \quad (\text{B.3})$$

$$\begin{aligned} &= \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int \frac{d\bar{\chi} d\Delta\chi}{\bar{\chi}^2} \int \frac{dk_1 dk_2}{(2\pi)^2} \tilde{K}_j^*(k_1) \tilde{K}_k(k_2) \\ &\times \int \frac{dk'}{2\pi} P(L/\bar{\chi}, k') e^{i(k-k')\Delta\chi} e^{-i(k_1 - k_2)\bar{\chi} - i(k_1 + k_2)\Delta\chi/2} \end{aligned} \quad (\text{B.4})$$

$$\begin{aligned} &= \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int \frac{d\bar{\chi}}{\bar{\chi}^2} \int \frac{dk_1 dk_2}{(2\pi)^2} \tilde{K}_j^*(k_1) \tilde{K}_k(k_2) \\ &\times \int \frac{dk'}{2\pi} P(L/\bar{\chi}, k') (2\pi) \delta(k - k' - \bar{k}) e^{-i(k_1 - k_2)\bar{\chi}} \quad (\bar{k} = \frac{k_1 + k_2}{2}) \end{aligned} \quad (\text{B.5})$$

$$\begin{aligned} &= \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int \frac{d\bar{\chi} d\bar{k}}{2\pi \bar{\chi}^2} P(L/\bar{\chi}, k - \bar{k}) \\ &\times \int \frac{d\Delta k}{2\pi} e^{-i(\Delta k)\bar{\chi}} \tilde{K}_j^*(\bar{k} + \Delta k/2) \tilde{K}_k(\bar{k} - \Delta k/2) \end{aligned} \quad (\text{B.6})$$

This handles the varying $\bar{\chi}$ problem but doesn't seem to be leading in a useful direction. The limit on the $\bar{\chi}$ integral is now tied up in the Fourier transforms, making it slightly harder to see what's going on.

Or we could try

$$C_\ell(k) = \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int d\chi_1 d\chi_2 K_j(\chi_1) K_k(\chi_2) e^{ik(\chi_1 - \chi_2)} C_L(\chi_1, \chi_2) \quad (\text{B.7})$$

$$\begin{aligned} &= \sum_{L\lambda} F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) \int \frac{dk' d\bar{\chi}}{2\pi \bar{\chi}^2} P(L/\bar{\chi}, k') \\ &\times \int d\Delta\chi K_j\left(\bar{\chi} + \frac{\Delta\chi}{2}\right) K_k\left(\bar{\chi} - \frac{\Delta\chi}{2}\right) e^{i(k-k')\Delta\chi} \end{aligned} \quad (\text{B.8})$$

Let us define (see also further discussion below)

$$W_{jk}(\bar{\chi}, k) \equiv \int dx K_j(\bar{\chi} + x/2) K_k(\bar{\chi} - x/2) e^{ikx} \quad (\text{B.9})$$

so that we can write

$$C_\ell(k) = \sum_{L\lambda} \int \frac{dk' d\bar{\chi}}{2\pi \bar{\chi}^2} P(L/\bar{\chi}, k') F_{\ell L\lambda} \sum_{jk} P_\lambda(\hat{n}_j \cdot \hat{n}_k) W_{jk}(\bar{\chi}, k - k') \quad (\text{B.10})$$

We see that $W_{jk}(\bar{\chi}, k)$ should have support only for a “thin” range of $\bar{\chi}$, and only for $k \approx 0$ if K_j is “broad and featureless”. These are nice properties. As in the main text, if W_{jk} is not rapidly varying in jk we expect the sum over λ to be restricted to very small λ . Can we efficiently evaluate/store this “transformation matrix”?

Note, we get back towards the earlier result if we notice that

$$W_{jk}(\bar{\chi}, k) = \int \frac{dk_1 dk_2}{(2\pi)^2} (2\pi) \delta^{(D)} \left(k - \frac{k_1 + k_2}{2} \right) \tilde{K}_j(k_1) \tilde{K}_k(k_2) e^{i(k_1 - k_2)\bar{\chi}} \quad (\text{B.11})$$

Move to $\bar{k}$ and Δk and you have the earlier result I think (up to a sign in the FT which I may have dropped). Maybe this helps us to rapidly compute W_{jk} ? I suspect if we went this Fourier route we’d likely want to apply some sort of apodization to our K_j functions, to avoid issues with trying to get a step function from a bunch of interfering FTs.

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